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ORIGINAL ARTICLE

Distribution of virulence factors and its relatedness towards the antimicrobial response of enterotoxigenic *Escherichia coli* strains isolated from patients in Kolkata, India

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Abstract

Aim: Enterotoxigenic *Escherichia coli* (ETEC) is one of the most widely recognized diarrhoeal pathogens in developing countries. The advancement of ETEC vaccine development depends on the antigenic determinants of the ETEC isolates from a particular geographical region. So, the aim here was to comprehend the distribution of virulence determinants of the clinical ETEC strains of this region. Additionally, an attempt was made to find any correlation with the antimicrobial response pattern.

Methods and results: Multiplex PCR was employed to identify virulence determinants followed by confirmatory singleplex PCR. For observation of antibiotic response, the Kirby-Bauer method was used. Out of 379 strains, 46% of strains harboured both the enterotoxins ST and LT, whereas 15% were LT only. Among the major colonization factors (CFs), CS6 (41%) was the most prevalent followed by CFA/I (35%) and CFA/III was the lowest (3%). Among the minor CFs, CS21 (25%) was most prevalent, while CS15 showed the lowest (3%) presence. Among the non-classical virulence factors, EatA (69%) was predominant. ETEC strains harbouring CS6 showed resistance towards the commonly used drug Ciprofloxacin (70%).

Conclusion: CS6 and *elt*+*est* toxin genes co-occurred covering 51% of the isolates. CS21 was found in most strains with *est* genes (43%). EatA was found to occur frequently when ST was present alone or with LT. CS6-harbouring strains showed an independent correlation to antimicrobial resistance.

Significance and Impact of the Study: This study would aid in identifying the commonly circulating ETEC isolates of Kolkata, India, and their prevalent virulence determinants. Knowledge of antibiotic resistance patterns would also help in the appropriate use of antibiotics. Furthermore, the study would aid in identifying the multivalent antigens suitable for region-specific ETEC vaccines with maximum coverage.

KEYWORDS

antimicrobial agents, colonization factor, Enterotoxigenic *Escherichia coli*, prevalence, virulence factors

INTRODUCTION

Enterotoxigenic *Escherichia coli* (ETEC) is an enteric pathogen estimated to cause about 400 million cases of diarrhoea and 157,000 deaths in children <5 years of age. Additionally, the bacteria also cause 44 million cases and 89,000 deaths in older children & adults. ETEC causes 8.5 million Disability Adjusted Life Years (DALYs) or 1 million Years Lived with Disability (YLDs) (Bourgeois, Wierzba & Walker, 2016). ETEC is the leading cause of diarrhoea in developing countries and among travellers from developed countries to these endemic regions (Arduino & Dupont, 1993). In developed nations like the United States, ETEC is acknowledged as a major cause of foodborne disease.

ETEC disease burden estimation of all ages has increased significantly in the number of deaths from 59,200 in 2013 to 74,100 in 2015 (Hosangadi, Smith, Kaslow, et al., 2019). Globally ETEC is detected frequently in Southeast Asian, African, and Middle East regions of the world. In Iraq, ETEC was detected in almost 75% of food items consumed by humans followed by diarrhoea during 2016–2017 (Taha & Yassin, 2019).

Global Enteric Multicentric Study (GEMS) indicated that ETEC was one of the four important pathogens related to diarrhoeal illnesses among children <5 years old in Africa and South Asia (Anderson IV et al., 2019). Although, it was previously thought that ETEC burden was lower, GEMS and MAL-ED molecular re-analysis data indicated the opposite spectrum. ETEC was among the top four pathogens causing diarrhoea-associated deaths among children <5 years old as estimated by CHERG (Hosangadi, Smith, & Giersing, 2019).

In the central region of Kenya, about 6%–7% of all diarrhoeal children were detected positive for ETEC (Mbuthia et al., 2018). In India, according to the global burden of diseases (GBD) 2015, around 6% of all diarrhoeal deaths in children aged less than 5 years were due to ETEC. In this eastern region of India, almost 5% of diarrheagenic *Escherichia coli* (DEC) was ETEC (Dutta et al., 2013). Out of the detected DEC, ETEC was the third most prevalent cause of pathogenesis (about 13.6%) between 2008 and 2012 in Karnataka, India (Singh et al., 2019). Surveillance study suggested that about 4% of clinical cases were ETEC among all diarrhoea infected patients hospitalized in Kolkata (Nair et al., 2010).

Pathogenesis of ETEC causes the release of electrolytes and water during watery diarrhoea. This was due to the release of plasmid-encoded two enterotoxins—heat-labile (LT) and heat-stable (ST). LT, encoded by *elt* gene, is an 84-kDa protein consists of an AB5 multimeric structure, where a pentamer of B chains has a membrane-binding function and the A chain is responsible for toxin activity. Structurally and functionally, LT is the mirror of cholera toxin. ST, encoded by *est* gene, is a non-antigenic low-molecular-weight peptide of 18–19 amino acids. It is of two variants, STp and STh, named after their discovery from porcine and humans respectively (Rao, 1985). ETEC strains produce either ST or LT or both; however, the ratio of different types of toxin-producing ETEC varies in different geographic regions (Qadri et al., 2005).

To release the enterotoxins, the bacteria first attach to the epithelium of the small intestine through different colonization factors (CF), also known as antigenic fimbriae called colonization factor antigens (CFA), a major virulence determinant for initiating pathogenesis (Gaastra & Svennerholm, 1996). CFAs encoded by ETEC are mainly divided into four major groups—CFA/I, CFA/II, CFA/III, and CFA/IV. These CFA groups are further subdivided according to their antigenic properties as coli surface antigens (CS). CFA/I is a single member of the first group. CFA/II group encodes CS3 alone or in combination with CS1 and CS2. CFA/III or CS8 is also a single member of this group. CFA/IV group encodes CS6 alone or in combination with CS4 and CS5. Rigid fimbriae CS1, CS2, CS4, CS5 and CFA/I are ~6 nm in diameter, whereas flexible fibrialle CS3 is ~3 nm in diameter (Vidal et al., 2019). Afimbrial CS6 is an assembly of two subunits, CssA and CssB associated tightly in an equal (1:1) stoichiometry (Sabui et al., 2016). Beside these main CF groups, there also exists minor CF which also helps in adherence of ETEC to the intestinal epithelium. Minor CFs include CS7, CS12, CS13, CS14, CS15, CS17, CS18, CS19, CS20, CS21, CS22, CS23, CS26, CS27, CS28, CS30 and PCF071 (Del Canto et al., 2012; Gaastra & Svennerholm, 1996; Mentzer et al., 2017). With the identification of CS30, there are 23 well-characterized CFs to date (Bhakat et al., 2018). Two more CFs are known in addition to these strains making the total number of CFs 25. The CFs together with the toxins are considered classical virulence factors. Out of the known CFs, only a few are prevalent such as CS6, CS5, CFA/I. The CFs like CS19, CS13, CS18 and CS15 are generally less common globally.

Besides the CFs, studies have indicated that other putative factors playing a role in ETEC pathogenesis, and these are known as 'non-classical virulence factors (NCVFs)'. EatA, EtpA, LeoA, TibA and Tia are the most commonly found NCVF in ETEC strains worldwide. EatA is a serine protease autotransporter that alleviates the delivery of LT toxin (Patel et al., 2004). EtpA is a secreted 170-kDa glycoprotein that plays a role in adherence of the bacteria to the intestinal mucosa (Fleckenstein et al., 2006). LeoA, a cytoplasmic protein, acts to maximize LT secretion and invasion of the epithelial cells by acting as a molecular bridge (Fleckenstein et al., 2000). TibA and Tia are the invasins which are autotransporters. They are predicted to promote adherence and promote the formation of biofilm in the host (Fleckenstein et al., 2010). More than 50% of the strains had EatA and EtpA as their NCVFs isolated from ETEC isolates in Bangladesh between 1998 and 2011 (Luo et al., 2015). In Bolivian children positive for ETEC, ClyA was the most commonly detected NCVF followed by EatA (Gonzales et al., 2013).

Until now, there is no effective licensed efficacious ETEC vaccine available and it allows indiscriminate use of antibiotics to curb the disease. As ETEC is prevalent throughout the globe and its variants are geo-region specific, so it is mandatory to know the disparity in the antibiotic response profile (Medina et al., 2015). About 20%–50% of all ETEC cases in developing countries emerge due to the wide range of antimicrobial resistance among the strains (Ericsson, 2003). Along with this, the emergence of multidrug-resistant ETEC is making the use of antibiotics unavailing against these pathogens (Diemert, 2006). Hence, unwrapping the antibiotic response of clinical ETEC isolates becomes a matter of immense importance.

According to WHO, the development of a practical and effective vaccine against ETEC is of high priority, especially for low- to middle-income countries which can be achieved by including prevalent toxins, CFs and NCVFs harboured by clinical strains. The strategy of developing a vaccine includes attenuated ETEC expressing CFs, multi-epitope fusion antigens, inactivated fimbriated ETEC, ST toxoids. As well as triggering intestinal secretory IgA antibodies to counter CFs and prevent ETEC from colonizing the digestive system is critical for effective immunization (Vidal et al., 2019).

The prevalence of these CFs and NCVFs is geographically based and changes with time. So, information about the prevalence of ETEC virulence determinants over a longer period of time is important for designing an effective ETEC vaccine with broader coverage. Therefore, monitoring the prevalence of various CFs and NCVFs changes with time turns out to be fundamental for defining the multivalent ETEC vaccine that would work over a wide scope of the topographical areas and effective for the long term.

In this study, the aim was to identify the commonly circulating virulence determinants of ETEC isolates from Kolkata, India. The antimicrobial response pattern among the strains having the most predominant virulence determinant was also studied. This study is expected to improve the knowledge about the strains from this region of India and their antimicrobial response pattern that would help to prescribe antibiotics for diarrhoeagenic patients with ETEC infection.

MATERIALS AND METHODS

Isolation and Identification of ETEC

In the hospital-based epidemiological surveillance study at the National Institute of Cholera and Enteric Diseases (NICED), stool specimens were collected from diarrhoeal patients admitted at the Infectious Diseases and Beliaghata General Hospital (ID & BG Hospital, Kolkata) and Dr. B.C. Roy Post Graduate Institute of Pediatric Sciences during 2015–2019. During this period, a total of 379 strains were positive for ETEC using the PCR-based analysis and collected for further investigation from the archive of NICED.

ETEC strains were plated onto MacConkey agar (SRL) plates and incubated for 16–18 h at 37°C. Colonies fermenting lactose on MacConkey agar plate having round and pink morphology were reconfirmed as ETEC and later used for experiments. These strains were further plated on Luria Bertini (LB) agar (Sigma Aldrich) plates. The culture from this nonselective medium was grown in LB and finally stored in 15% glycerol at −80°C for further use.

Extraction of DNA templates

DNA templates were prepared by the boil lysis method. Few colonies of ETEC from agar plate were taken in Tris-EDTA buffer (pH-6.8) and boiled for 10 min followed by immediate cooling in ice. Then the mix was centrifuged and an aliquot of the supernatant was used as a DNA template for PCR.

Detection of virulence determinants by multiplex PCR

For molecular detection of virulence determinants, a total of three primer sets for toxin detection, 24 primer sets for CFs and five primer sets for Non-classical virulence determinants were used (The details of primers are given in Table S1). The PCR reaction was carried in a total volume of 20 μl under the following condition-denaturation at 95°C for 10 min, denaturation at 94°C for 1 min, annealing at 52°C for 30 s, extension at 72°C for 60 s/kb of amplicon and a final extension at 72°C for 10 min. The virulence determinants tested positive in multiplex PCR was confirmed by simplex PCR under the same condition using respective primers for each of the genes.

Antimicrobial susceptibility test

The Kirby–Bauer method was used for bacterial susceptibility towards antimicrobial agents. For the antimicrobial susceptibility test, representative isolates were selected on the basis of having predominant CF. The antibiotic discs (Becton, Dickinson and Company) used are as follows: Imipenem (Ipm)-10 μg; Chloramphenicol (C)-30 μg; Cefapime (Fep)-30 μg; Doxycycline (D)-30 μg; Ceftriaxone (Cro)-30 μg; Tetracycline (Te)-30 μg; Norfloxacin (Nor)-10 μg; Sulfamethoxazole W/ Trim (Sxt); Ciprofloxacin (Cip)-5 μg; Ampicillin (Am)-10 μg; Azithromycin (Azm)-15 μg; Nalidixic Acid (Na)-30 μg; Erythromycin (E)-15 μg; Streptomycin (S)-10 μg.

The test was performed on the Mueller–Hinton (MH) agar plates following Clinical and Laboratory Standards Institute (CLSI) guidelines (Hsueh et al., 2010). The isolates were inoculated in MH Broth and incubated at 37°C up to 0.5 McFarland standard turbidity. The culture was streaked thoroughly onto MH Agar plates using cotton swabs. After air-drying the plates, antibiotic discs were placed and after overnight incubation at 37°C, the zone of inhibition was measured as per the manufacturer's protocols.

Statistical analysis

For comparison of two variables, a chi-square test was used. A $p < 0.05$ was considered statistically significant.

RESULTS

Toxin types of ETEC isolates

Archived isolates of ETEC were analysed in this study. Of 379 ETEC isolates, 38% ($n = 142$) strains were yielded along with other pathogens and 62% ($n = 237$) strains were isolated where ETEC was the sole pathogen. All the 379 strains screened were positive for toxin genes. Among the ETEC isolates, 175 strains harboured both *elt* + esth toxin genes (46%) followed by the strains harbouring *esth* gene alone (37%) and strains harbouring the only *elt* as toxin gene (15%). Only 1% of strains harboured *elt* + estp toxin gene and the frequency of *estp* only strains was also 1%. Only one strain was detected positive for the presence of *elt* + esth + *estp* (Table 1).

Distribution of toxin genes among ETEC isolates identified over time

ETEC strains isolated over 2015–2019 showed that the cases of ETEC infection were highest in 2015 (Table 1). The cases gradually decreased over the next 4 years. In all these years, *elt*+esth was the most frequently detected toxin gene combination among the ETEC isolates followed by *esth* only strains, making the *elt*+esth toxin gene combination the most prevalent toxin type in this region of India. The *elt* gene toxin was detected almost three times more in 2016 ($n = 34$) than in 2015 ($n = 10$). Among the isolates during these 5 years, the presence of *estp*+elt was observed in alternate years, that is, in 2015, 2017 and 2019. The *estp* harbouring strains were also detected in these above-mentioned years only (Table 1).

Age-wise distribution of toxin genes among ETEC isolates

Among the archived isolates, 49% of ETEC strains were isolated from patients younger than 5 years of age and 45% were above the age of 18 years (Table 2). The rest 6% were in between the age of 5–18 years.

When the distribution of toxin genes was studied in these three different age groups, that is, <5 years, 5–18 years, and >18 years of age, it was found that in all age groups, *elt*+est harbouring strains were more than *elt* only or *est* only ETEC isolates (Table 2). In age groups <5 years *elt+est* harbouring strains were 43%, in the age group 5–18 years 48% and of isolates from adults it was 51% whereas 18%, 9% and 11% of strains harboured *elt* gene respectively. Four ETEC isolates from children of ≤5 year's age had both *elt+ estp* as toxin gene and this combination has not been found in patients of more than 5 years of age. Two strains in the age group 5–18 years were found harbouring e*stp* gene only. Only 1 ETEC isolate harbouring *elt+ esth+ estp was* found in the patient of 5–18 years of age group. However, when analysed it was observed that distribution of toxin genes is not statistically significantly associated with age groups ($p = 0.357$).

Level of dehydration in comparison with toxin types

When recorded data of ETEC patients were collected on the level of dehydration they had during infection, it was observed

TABLE 1 ETEC distribution over the years with toxin genes

Year	Toxin genes						Total
	elt	*est*	*elt+esth*	*elt+estp*	*estp*	*elt+esth+estp*	
2015	10	36	56	2	2	0	106
2016	34	33	34	0	0	0	101
2017	2	25	38	1	1	1	68
2018	5	22	27	0	0	0	54
2019	5	23	20	1	1	0	50
Total	56	139	175	4	4	1	379

Toxin genes were detected using PCR method. Please see Material and Methods for details.

elt—heat labile toxin gene; *esth*—heat stable toxin gene; human variant; *estp*—heat stable toxin gene; porcine variant.

TABLE 2 Occurrence of ETEC in different age group with toxin genes

Age	Toxin genes			p value
	elt (%)	*est* (%)	*elt*+*est* (%)	
0–5 ($n = 185$)	33(17.8)	72(38.4)	80(43.2)	0.357
>5–18 ($n = 23$)	2(8.0)	10(43.4)	11(47.8)	
18+ ($n = 171$)	21(12.2)	61(35.6)	89(52.0)	

Toxin genes were detected using PCR method. Please see Material and Methods for details.

elt—heat labile toxin gene; *est*,—heat stable toxin gene; includes both human (*esth*) and porcine (*estp*) variant.

TABLE 3 Dehydration level associated with toxin genes

Dehydration level	Toxin genes			Total	p value
	elt	*est*	*elt*+*est*		
None	14	31	31	76	0.687
Some	37	101	134	272	
Severe	4	11	16	31	

elt—heat labile toxin gene; est—heat stable toxin gene; includes both human (*esth*) and porcine (*estp*) variant.

that 72% of patients with some dehydration were maximum in number (Table 3). Severe dehydration was observed in the least number of patients (8%). The rest of the 20% of patients had not experienced any dehydration during the infection.

When this data of level of dehydration was compared to the type of toxin produced by ETEC strains, it was revealed that among *elt*+est harbouring ETEC strains, 74% were from patients having 'some' level of dehydration followed by 17% patients with no dehydration and only 9% experienced 'severe' level of dehydration. In *elt*, *est* and *elt*+*est* harbouring strains similar patterns of dehydration levels were observed with a maximum number of isolates with some level of dehydration and the least with a 'severe' level of dehydration. Statistical analysis revealed that there is no significant association between the level of dehydration and toxin genes ($p > 0.05$). The absence of association may be due to the small sample size and unavailability of data about if the patients had taken any over the counter drug before admission to the hospital.

Genotypic distribution of ETEC virulence factors

Among the ETEC isolates, both the classical CF and NCVF were present in 69% of the ETEC strains. Only CF genes were detected in 22% of the strains. Six percent of strains showed the presence of NCVF genes alone as their virulence determinants. No virulence determinants could be detected in 3% of the ETEC isolates in this study (Figure 1a).

The distribution of virulence factors over the years followed a similar pattern with isolates harbouring both CF+NCVF as their virulence determinants and least being the isolates without any detectable virulence factors (Figure 1b).

When the distribution of virulence factors in ETEC isolates was compared to the presence of toxin genes in the isolates, it was revealed that among 46% of strains harbouring CFs had both the *elt*+est toxin genes, followed by 40% of strains having *est* toxin gene. Distribution of the toxin genes among NCVF harbouring ETEC isolates revealed that maximum strains (47%) had both the *elt*+est toxin genes (Table 4).

Distribution of major CFs in ETEC isolates

The application of PCR enabled us to detect the presence of different CFs in this study as well as the prevalence of those CFs among ETEC isolates. Eight major CFs were targeted for detection grouped in CFA/I, CFA/II, CFA/III and CFA/IV (Table 5). The CFA/IV group was detected in 43% of the ETEC isolates followed by the CFA/I, and CFA/II group. The CFA/III was detected in the least number of isolates.

Major CF distribution in relation to toxin genes

Within the CFA/IV group, comprising of CFs CS4, CS5 and CS6, the CS6 was maximally detected (Table 6). CS6 harbouring ETEC isolates were detected in 41% of LT+ST strains. Among both LT-only and ST-only strains, CS6 detected almost in equal frequency, 34% and 36% respectively. Alongside CS6, in ST-only isolates, CFA/I was detected in almost the same number (37%) but in LT+ST strains it was detected in only 28% strains compared to 41% strains positive for CS6 (Table 6). Following CS6 and CFA/I, CS5 is another major CF found commonly in 21% CF positive ETEC isolates. The most prevalent combination of Major CF was found to be CS6 with CS5 followed by CFA/I with minor CF CS21. Among the prevalent major CF combination, 61% of CS6+CS5 ETEC isolates were found to harbour *elt*+*est* as their toxin gene followed by ST-only CS6+CS5 isolates (27%) and only 10% of CS6+CS5 isolate harboured *elt* as their toxin

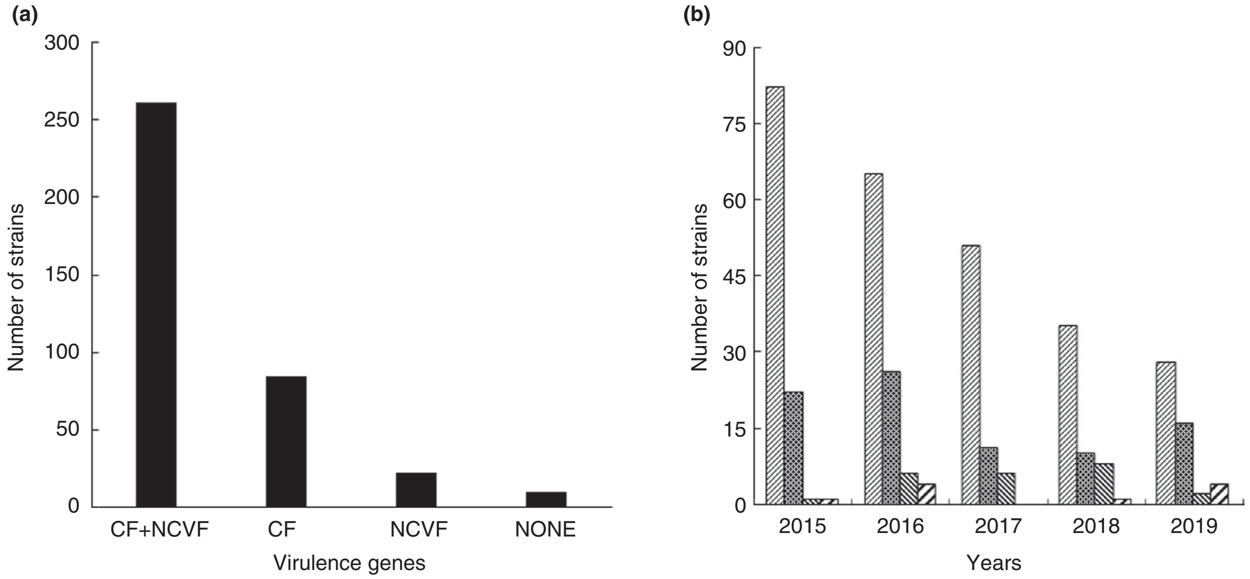

FIGURE 1 Distribution of ETEC virulence factors during the study period (2015–2019). (a) Shows overall distribution of virulence determinants. (b) (▩) CFs; (▧) NCVF; (▨) CF+NCVF; and (◪) None; represents year wise distribution of virulence determinants. A total of 379 clinical samples were screened for CFs—classical colonization factors; and NCVFs— NCVFs. CFs denote the isolates positive only for the classical colonization factors; NCVFs represents strains positive only for the NCVFs; CF+NCVF represents strains positive for both the virulence factors; and NONE represents strains negative for both the virulence factors. We tested a total of 29 virulence factors (24 CFs and 5 NCVFs) and the results shown are based on the screened factors

TABLE 4 ETEC Virulence factors distribution with toxin genes

Virulence factors	Toxin genes			
	elt	*est*	*elt+est*	Total
CF	48	143	163	354
NCVF	31	119	134	284
None	2	2	6	10

est includes both human (*esth*) and porcine (*estp*) variant.

Classical colonization factor (CF) genes studied here by PCR; please see Materials and Methods for details.

Non-classical virulence factor (VF) genes studied here by PCR; please see Materials and Methods for details.

No tested virulence factor detected.

TABLE 5 ETEC CF distribution with toxin genes

Major CF groups	Toxin genes			
	elt	*est*	*elt+est*	Total
I	19	52	51	122
II	6	22	55	83
III	4	3	3	10
IV	23	59	81	163

est includes both human (*esth*) and porcine (*estp*) variant.

Classical colonization factor (CF) genes, Classical CF genes studied here by PCR; please see Materials and Methods for details.

gene. Whereas 55% of CS6+CFA/I isolate harboured *elt+est* as their toxin gene. In CFA/I+CS21 isolates, 51% of strains harboured *elt+est* as prevalent toxin gene (Table 7).

TABLE 6 Prevalent CF distribution with toxin genes

Prevalent major CFs	Toxin genes			
	elt	*Est*	*elt+est*	Total
CS6	19	51	73	143
CFA/I	19	52	51	122
CS5	9	21	45	75

est includes both human (*esth*) and porcine (*estp*) variant.

Classical colonization factor (CF) genes, Classical CF genes studied here by PCR; please see Materials and Methods for details.

TABLE 7 Occurrence of the most prevalent combination of CF with toxin genes

Classical genes	Toxin			
	elt	*est*	*elt+est*	Total
CS6+CS5	7	18	40	65
CFA/I+CS21	3	17	21	41
CS6+CFA/I	3	13	20	36
CS6+CS21	0	8	6	14

est heat stable toxin gene; includes both *esth* and *estp*.

Distribution of major CFs over the years

During 2015–2019, CS6 was detected as the predominant CF over other CFs in every year, for example, 35% isolates in 2015, 44% isolates in 2016, 43% isolates in 2017, 24%

Journal of Applied Microbiology sfam

TABLE 8 Major CF distribution over the years

Major CF groups	CF	Years 2015	2016	2017	2018	2019	Total	Percent
CFA/I	CFA/I	32	42	16	17	15	122	34
CFA/II	CS1	5	17	6	3	6	37	10
	CS2	8	5	4	3	4	24	7
	CS3	14	24	12	5	7	62	18
CFA/III	CFA/III	1	3	4	1	1	10	3
CFA/IV	CS4	3	3	2	4	1	13	4
	CS5	22	25	16	7	5	75	21
	CS6	37	44	29	16	17	143	40

Classical colonization factor (CF) genes, Classical CF genes studied here by PCR; please see Materials and methods for details.

isolates in 2018, and 34% isolates in 2019 (Table 8). The detection of CFA/I in the ETEC isolates in these years were after CS6. In 2016, CS6 and CFA/I were detected almost in similar abundance in the ETEC isolates. In 2016 and 2017, CS6 was detected in a higher percentage (44% and 43% respectively) than that of 2015, 2018, and 2019. Among the rarely encountered CFs, CFA/III was detected in 3% of isolates only.

Minor CF among ETEC isolates

Seventy-five percent of the total ETEC strains were positive for harbouring at least one minor CF. Among these, 11%strains harboured *elt* toxin gene only, 42% harboured est toxin gene only and 47% harboured *elt*+est toxin gene. Among the minor CFs, CS21 was accounted for the highest presence (31%). In these strains, *elt*+est positive CS21 was maximum, whereas only three strains had the *elt* gene.

During the 2015–2019 period, CS21 was detected maximally over other minor CFs in every year, for example, 21% isolates in 2015, 19% isolates in 2016, 30% isolates in 2017, 27% isolates in 2018, and 18% isolates in 2019 (Table 9), This was followed by detection of CS23 in these years except in 2019, when no strains were found having CS23. In minor CFs, CS15 was found in the least number of strains found only. Four minor CFs, CS7, CS19, CS13, CS26 and CS30 were not detected in any of the ETEC isolates in our study.

Genotypic distribution of NCVF genes in ETEC isolates in the present study

The non-classical VFs were screened and found to be present in 75% of the ETEC strains during the study period 2015–2019. Out of these positive NCVFs strains, EatA (69%) was most commonly detected followed by EtpA (42%), tibA (18%), and tia (12%) (Table 10). LeoA gene (7%), was least detected NCVF in our study. Most of the NCVF harbouring ETEC isolates were in co-presence with at least one or more classical CFs. In this study, only 8% of the ETEC isolates had NCVF genes alone. EatA was mostly detected (42%) along with prevalent classical CF CS6. Six percent of EatA gene was detected in ETEC isolates having no detectable CFs. Out of all the ETEC isolates, 25% isolates were NCVF negative.

Distribution of NCVF genes in ETEC isolates in relation to toxin genes detected

When analysed in comparison with toxin genes, it was observed that *elt*+est-ETEC isolates harboured maximum NCVF in 47% strains (47%). In *elt* only strains, least number of NCVF (11%) was found. EatA, the predominant NCVF, found maximally in *elt*+est- strains (49%), and *elt* only strains, EatA was minimum (9%). Following EatA, EtpA also showed the same pattern of distribution accounting for half of the EtpA strains (50%) produced LT+ST toxins and almost equal percentage produced ST toxins (Table 10).

Year-wise distribution of NCVF genes among ETEC isolates

During the span of study from 2015 to 2019, EatA was the prevalent non-classical VF detected in our region followed by EtpA as the next common one. EatA was identified in the highest number in 2015, whereas EtpA was most detected in 2016. In 2016, the least common NCVF, LeoA was identified only in one strain. NCVF genes were found to be an independent variable over time (Table 11).

TABLE 9 Minor CF distribution over the years

Minor CFs	Years 2015	2016	2017	2018	2019	Total	Percent
CS21	23	20	21	15	9	88	25
CS23	20	8	6	3	0	37	10
CS20	18	0	10	0	3	31	9
CS28	17	1	3	2	3	26	8
CS27	12	6	0	1	1	20	6
PCF071	4	5	7	3	1	20	6
CS22	13	0	0	1	0	14	4
CS12	8	2	1	0	0	11	3
CS18	2	2	7	0	0	11	3
CS14	7	2	1	0	2	12	3
CS17	6	5	2	0	2	15	4
CS15	4	0	0	3	5	12	3
CS19	0	0	0	0	0	0	0
CS13	0	0	0	0	0	0	0
CS26	0	0	0	0	0	0	0
CS30	0	0	0	0	0	0	0

Classical colonization factor (CF) genes, Classical CF genes studied here by PCR; please see Materials and methods for details.

TABLE 10 Occurrence of ETEC NCVFs with toxin genes

NCVF	Toxin genes *elt*	*est*	*elt+est*	Total
EatA	17	82	97	196
EtpA	10	50	59	119
Tib	5	21	24	50
Tia	10	12	13	35
LeoA	3	8	9	20

est includes both human (*esth*) and porcine (*estp*) variant.

Non-classical virulence factor (VF) genes. Non-classical VF genes studied here by PCR; please see Materials and methods for details.

Antibiotic susceptibility of ETEC isolates

From this study, CS6 was identified as the prevalent CF during 2015–2019. The antibiotic susceptibility in the CS6-harbouring ETEC isolates were tested for antibiotic response against 14 antimicrobial agents (Table 12). All CS6-harbouring isolates were resistant to one or more antimicrobial agents with the most frequent resistance found against 10 μg streptomycin (99.5%) and 15 μg erythromycin (97.7%). More than 80% of strains were resistant to 30 μg Nalidixic Acid and 15 μg Azithromycin. All the ETEC isolates were sensitive to 10 μg Imipenem. The majority of the strains were sensitive to 30 μg Chloramphenicol. When combinations of different antibiotic resistance were analysed compared to the toxin types in CS6-harbouring strains, a very few strains showed a similar pattern of resistance to different antibiotic combinations (Table S3).

To compare the results, we also tested 10 strains, in which no known virulence factors were detected. Among these isolates, almost a similar pattern of response to antimicrobial agents was found (Table 12).

DISCUSSION

Examining the presence of toxin genes in clinical ETEC strains during this study period, it was observed that the *elt+est* harbouring ETEC strains were most common, and *elt* only strains were found in the lowest number. This result is comparable with the findings from studies conducted in Bangladesh (Begum et al., 2014). Findings from a previous study in this region of Kolkata also indicated the same pattern of the presence of toxin genes in ETEC isolates (Bhakat et al., 2018). Global enteric multicenter study (GEMS) on Asia and Africa reported 68% were either ST only or LT/ST ETEC strains (Vidal et al., 2019). In contrast, the global prevalence of *elt* toxin gene is much higher than *est*, either alone or together with *elt* (*est+elt*). A study revealed that LT toxin genes are present in approximately 60% of the field ETEC strains associated with diarrheal incidents in humans, either LT alone (27%) or in combination with ST (33%) (Isidean et al., 2011).

Results presented here trends for the presence of at least one CF are present in 91% of ETEC isolates. This can be

TABLE 11 NCVF distribution with time

NCVF	Years 2015	2016	2017	2018	2019	Total	Percent
EatA	64	46	41	29	16	196	69
EtpA	23	41	24	15	16	119	42
TibA	19	11	13	5	2	50	18
Tia	10	7	7	6	5	35	12
LeoA	12	1	4	2	1	20	7

Non-classical virulence factor (VF) genes. Non-classical VF genes studied here by PCR; please see Materials and methods for details.

TABLE 12 Response of CS6 harbouring ETEC isolates and ETEC isolates having no detectable virulence factors to antimicrobial agents

Antibiotics	Number of CS6 resistant strains (%)	Number of none resistant strains (%)
Imipenem (Ipm)—10 µg	0 (0)	0 (0)
Chloramphenicol (C)—30 µg	8 (6)	2(20)
Cefapime (Fep)—30 µg	34 (24)	3 (30)
Doxycycline (D)—30 µg	39 (27)	2 (20)
Ceftriaxone (Cro)—30 µg	46 (32)	2 (20)
Tetracycline (Te)—30µg	47 (33)	2 (20)
Norfloxacin (Nor)—10µg	51 (36)	5 (50)
Sulfamethoxalone W/Trim (Sxt)	58 (41)	3(30)
Ciprofloxacin (Cip)—5 µg	100 (70)	6 (60)
Ampicillin (Am)—10 µg	107 (75)	8 (80)
Azithromycin (Azm)—15 µg	120 (84)	7 (70)
Nalidixic Acid (Na)—30 µg	124 (87)	10 (100)
Erythromycine (E)—15 µg	140 (98)	10(100)
Streptomycine (S)—10 µg	142 (99)	10 (100)

None refers to those ETEC isolates in which no virulence factors were detected by our tested virulence factors by PCR method.
Antibiotic disks were used (BD, USA) for this study using Kirby–Bauer Disc diffusion method.

compared to the frequencies of 23–94% isolates positive for CF reported in a systematic review done by Isidean et al., 2011. In Bangladesh, 49% of the ETEC isolates were positive for at least one CF (Begum et al., 2014). A study on Nicaraguan children reported that at least 50% of ETEC strains were positive for one CF (Vilchez et al., 2014). In Shenzhen, China, 54% of ETEC isolates had one or more CF present in ETEC strains (Li et al., 2017). The high presence of CF in ETEC isolates in our study (Table S2) was most probably due to the inclusion of 24 CFs in the detection protocol.

During an earlier analysis, findings from Kolkata strains between 2008 and 2014 showed CS21 and CS6 were the predominant CFs (Bhakat et al., 2018). In this study, the distribution of CS6 is predominant followed by CFA/I, rather than CS21. A similar trend in the presence of CS6 was also observed in China and Guatemala (Li et al., 2017; Torres et al., 2014). In the Kolkata region of India, CS6 was mostly detected in the clinical ETEC isolates (Dutta et al., 2013; Ghosal et al., 2007). The CS6 was the predominant CF in ETEC isolates from Zambia (Simuyandi et al., 2019), Guatemala (Torres et al., 2014). A study in Bali, Indonesia indicated CFA/I as the predominant CF (Subekti et al., 2003). Our revelations are also in accordance with GEMS discovery from studies conducted on Asian and African countries where CFA/I as well as CS1-CS6 are some major CF antigens. In rural Egypt, most of the ETEC strains had CFA/I, followed by CS6 as the predominant CF (Shaheen et al., 2004). CFA/I and CS17 are the most common virulence determinant detected in Bolivia (Rodas et al., 2011). In accordance with our findings, a study on the ETEC population in Nepal revealed that CS21 (62.6%) and CS6 (30.2%) were the most prevalent CFs (Margulieux et al., 2018). Another study in our laboratory earlier detected CS21 as the prevalent CF followed by CS6 among clinical ETEC isolates from Kolkata (Bhakat et al., 2018). In Bangladesh, frequently detected CFs were

CS5, CS6 and CS1 (Begum et al., 2014). The co-occurrence of CS5 and CS6 in ETEC strains was found in our study. This trend was also similar to the previous study (Bhakat et al., 2018). This is in accordance with the study that stated CS6 is expressed alone or in combination with CS5 or CS4 (Gaastra & Svennerholm, 1996; Wolf, 1997). This study found only three ETEC strains showing co-occurrence of CS6 and CS4.

In our study, we detected most of the ETEC strains having 1–5 CFs except a few strains having more than the average number of CFs (Table S2). So, there might be a possibility of having mixed strains in archived stock. This might suggest that during stock preparation of archived strains there could be some technical limitations which might give rise to mixed strains.

Analysis with respect to the presence of non-classical virulence determinants showed that three-fourth of the ETEC strains contained at least one or more of these factors. Among these, EatA was the most predominant one followed by EtpA. A study on ETEC clinical isolates of Northern Colombia, South America also indicated EatA as one of the most prevalent NCVF detected (Guerra et al., 2014). EatA was generally identified in strains having the presence of CFs, recommending a significant part to advance intestinal colonization. A similar trend of the presence of EatA and CFs were also found in Chilean ETEC isolates (Del Canto et al., 2011). However, any noteworthy connection of conjunction of EtpA and CFs was not found. This might be because of the cooperation of EtpA with highly conserved flagellin and accordingly elevating adherence to the intestinal wall (Roy et al., 2009). The other three NCVFs Tia, TibA and LeoA were found in low recurrence. Among these, LeoA was found in the most minimal recurrence and two strains with LeoA were negative for any classical CF genes. This outcome repudiates with the examination from Chile, Colombia and Guatemala where the greater part of the LeoA strains were negative for any classical genes (Del Canto et al., 2011; Guerra et al., 2014; Torres et al., 2014).

Among the remaining one-fourth of strains negative for NCVFs, only 10 strains were negative for known CFs. There were only 4% of CF negative strains that were positive for NCVF, that is: Tia, TibA and EtpA. So, despite the inclusion of most of the discovered CFs and NCVFs, we were unable to detect any factor responsible for colonization in 10 strains suggesting that there may be additional colonization determinants yet to be perceived.

Globally, reports of ETEC isolates negative for any virulence factors other than toxins are common. In Bangladesh, 51% of ETEC strains were negative for any CFs (Begum et al., 2014). In Thailand, 41% of ETEC isolates were negative for CFs (Puiprom et al., 2010). Similarly, in 46% of strains, no CFs were detected in Shenzen, China, and 33% in Iranian children (Li et al., 2017; Nazarian et al., 2014). This study identified just 3% of ETEC isolates negative for any virulence determinants. This was possible with the inclusion of more CF in the study design. This is an indication that for maximum coverage on the epidemiological survey we should follow the broad consideration of epidemiologic data about ETEC Classical and Non-Classical virulence determinants. This should be an important aspect in tracking the diversity of ETEC isolates in this region of India and prioritize the choice of the potential vaccine candidate, explicitly for this geographic realm of India.

This study on the antimicrobial response pattern of CS6-harbouring ETEC isolates towards common antimicrobial agents used for the treatment of ETEC-related diarrhoea, demonstrated that almost all these strains were resistant to early-generation drug erythromycin and streptomycin. More than 70% of CS6 harbouring ETEC isolates were ampicillin-resistant which is comparable to maximum resistance to ampicillin in Bolivian children (Rodas et al., 2011). A similar observation was reported in a study on Peruvian children that showed 64% of ETEC strains were resistant to ampicillin (Medina et al., 2015). The results obtained here are in accordance with a previous study in India where ETEC resistance to ciprofloxacin was reported (Chakraborty et al., 2001). In this investigation, it was observed that more than 60% of CS6-harbouring ETEC isolates of this region showed resistance towards nalidixic acid, azithromycin and tetracycline which were also observed in ETEC strains of Bangladesh showing similar resistance to the above antibiotics (Rahman et al., 2020). A study in Eastern Odisha, India revealed that more than 50% of ETEC strains were resistant to ceftriaxone and 2.4% of those isolates were resistant to imipenem (Moharana et al., 2019), whereas our investigation showed only 6% strains were resistant to ceftriaxone and no strains were resistant to imipenem. In this study, the absence of major similarity in resistance towards different combinations of antibiotics shows that there is a huge genetic diversity and acquiring pattern of resistance is also very diverse among the CS6-positive isolates.

WHO recommended using the prevalent CF around the world which includes CFA/I, CS1-CS6 along with LT-B of ETEC for developing vaccine candidates (Bourgeois et al., 2016). From our examination, it may be presumed that strains holding both the enterotoxins LT and ST, alongside classical CF CS6, CS5, CFA/I and CS21 along with NCVF EatA are the probable combination of virulence determinates coursing in this region.

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CONFLICT OF INTEREST

The authors have no conflict of interest to declare.

AUTHOR CONTRIBUTIONS

IM, DB and NSC conceptualized the study. IM, DB and GC performed experiments. IM, DB, AKM and NSC prepared the original draft of the manuscript. AM, SS, AKD and AKM collaborated in this study. All authors did data analysis, draft review, editing and approval.

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REFERENCES

Anderson, J.D., Bagamian, K.H., Muhib, F., Amaya, M.P., Laytner, L.A., Wierzba, T. et al. (2019) Burden of enterotoxigenic *Escherichia coli* and shigella non-fatal diarrhoeal infections in 79 low-income and lower middle-income countries: a modelling analysis. *The Lancet Global Health*, 7(3), e321–e330.

Arduino, R.C. & Dupont, H.L. (1993) Travellers' diarrhoea. *Baillière's Clinical Gastroenterology*, 7(2), 365–385.

Begum, Y.A., Baby, N.I., Faruque, A.S.G., Jahan, N., Cravioto, A., Svennerholm, A.-M. et al. (2014) Shift in phenotypic characteristics of enterotoxigenic *Escherichia coli* (ETEC) isolated from diarrheal patients in Bangladesh. *PLoS Neglected Tropical Diseases*, 8(7), e3031.

Bhakat, D., Debnath, A., Naik, R., Chowdhury, G., Deb, A.K., Mukhopadhyay, A.K. et al. (2018) Identification of common virulence factors present in enterotoxigenic *Escherichia coli* isolated from diarrhoeal patients in Kolkata, India. *Journal of Applied Microbiology*, 126(1), 255–265.

Bourgeois, A.L., Wierzba, T.F. & Walker, R.I. (2016) Status of vaccine research and development for enterotoxigenic *Escherichia coli*. *Vaccine*, 34(26), 2880–2886.

Chakraborty, S., Deokule, J.S., Garg, P., Bhattacharya, S.K., Nandy, R.K., Nair, G.B. et al. (2001) Concomitant infection of enterotoxigenic *Escherichia coli* in an outbreak of cholera caused by *Vibrio cholerae* O1 and O139 in Ahmedabad, India. *Journal of Clinical Microbiology*, 39(9), 3241–3246.

Del Canto, F., Botkin, D.J., Valenzuela, P., Popov, V., Ruiz-Perez, F., Nataro, J.P. et al. (2012) Identification of coli surface antigen 23, a novel adhesin of enterotoxigenic *Escherichia coli*. *Infection and Immunity*, 80(8), 2791–2801.

Del Canto, F., Valenzuela, P., Cantero, L., Bronstein, J., Blanco, J.E., Blanco, J. et al. (2011) Distribution of classical and nonclassical virulence genes in enterotoxigenic *Escherichia coli* Isolates from Chilean Children and tRNA gene screening for putative insertion sites for genomic Islands. *Journal of Clinical Microbiology*, 49(9), 3198–3203.

Diemert, D.J. (2006) Prevention and self-treatment of traveler's diarrhea. *Clinical Microbiology Reviews*, 19(3), 583–594.

Dutta, S., Guin, S., Ghosh, S., Pazhani, G.P., Rajendran, K., Bhattacharya, M.K. et al. (2013) Trends in the prevalence of diarrheagenic *Escherichia coli* among hospitalized diarrheal patients in Kolkata, India. *PLoS One*, 8(2), e56068.

Ericsson, C.D. (2003) Travellers' diarrhoea. *International Journal of Antimicrobial Agents*, 21(2), 116–124.

Fleckenstein, J.M., Hardwidge, P.R., Munson, G.P., Rasko, D.A., Sommerfelt, H. & Steinsland, H. (2010) Molecular mechanisms of enterotoxigenic *Escherichia coli* infection. *Microbes and Infection*, 12(2), 89–98.

Fleckenstein, J.M., Lindler, L.E., Elsinghorst, E.A. & Dale, J.B. (2000) Identification of a gene within a pathogenicity island of enterotoxigenic *Escherichia coli* H10407 required for maximal secretion of the heat-labile enterotoxin. *Infection and Immunity*, 68(5), 2766–2774.

Fleckenstein, J.M., Roy, K., Fischer, J.F. & Burkitt, M. (2006) Identification of a two-partner secretion locus of enterotoxigenic *Escherichia coli*. *Infection and Immunity*, 74(4), 2245–2258.

Gaastra, W. & Svennerholm, A.M. (1996) Colonization factors of human enterotoxigenic *Escherichia coli* (ETEC). *Trends in Microbiology*, 4(11), 444–452.

Ghosal, A., Bhowmick, R., Nandy, R.K., Ramamurthy, T. & Chatterjee, N.S. (2007) PCR-based identification of common colonization factor antigens of enterotoxigenic *Escherichia coli*. *Journal of Clinical Microbiology*, 45(9), 3068–3071.

Gonzales, L., Sanchez, S., Zambrana, S., Iniguez, V., Wiklund, G., Svennerholm, A.-M. et al. (2013) Molecular characterization of enterotoxigenic *Escherichia coli* isolates recovered from children with diarrhea during a 4-year period (2007 to 2010) in Bolivia. *Journal of Clinical Microbiology*, 51(4), 1219–1225.

Guerra, J.A., Romero-Herazo, Y.C., Arzuza, O. & Gómez-Duarte, O.G. (2014) Phenotypic and genotypic characterization of enterotoxigenic *Escherichia coli* clinical isolates from Northern Colombia, South America. *BioMed Research International*, 2014, 1–11.

Hosangadi, D., Smith, P.G. & Giersing, B.K. (2019) Considerations for using ETEC and Shigella disease burden estimates to guide vaccine development strategy. *Vaccine*, 37(50), 7372–7380.

Hosangadi, D., Smith, P.G., Kaslow, D.C. & Giersing, B.K. (2019) WHO consultation on ETEC and Shigella burden of disease, Geneva, 6–7th April 2017: meeting report. *Vaccine*, 37(50), 7381–7390.

Hsueh, P.R., Ko, W.C., Wu, J.-J., Lu, J.-J., Wang, F.-D., Wu, H.-Y. et al. (2010) Consensus statement on the adherence to Clinical and Laboratory Standards Institute (CLSI) Antimicrobial Susceptibility Testing Guidelines (CLSI-2010 and CLSI-2010-update) for Enterobacteriaceae in Clinical Microbiology Laboratories in Taiwan. *Journal of Microbiology, Immunology, and Infection*, 43(5), 452–455.

Isidean, S.D., Riddle, M.S., Savarino, S.J. & Porter, C.K. (2011) A systematic review of ETEC epidemiology focusing on colonization factor and toxin expression. *Vaccine*, 29(37), 6167–6178.

Li, Y., Luo, Q., Shi, X., Lin, Y., Qiu, Y., Lv, D. et al. (2017) Phenotypic and genotypic characterization of clinical Enterotoxigenic *Escherichia coli* isolates from Shenzhen, China. *Foodborne Pathogens and Disease*, 14(6), 333–340.

Luo, Q., Qadri, F., Kansal, R., Rasko, D.A., Sheikh, A. & Fleckenstein, J.M. (2015) Conservation and immunogenicity of novel antigens in diverse isolates of enterotoxigenic *Escherichia coli*. *PLoS Neglected Tropical Diseases*, 9(1), e0003446.

Margulieux, K.R., Srijan, A., Ruekit, S., Nobthai, P., Poramathikul, K., Pandey, P. et al. (2018) Extended-spectrum β-lactamase prevalence and virulence factor characterization of enterotoxigenic *Escherichia coli* responsible for acute diarrhea in Nepal from 2001 to 2016. *Antimicrobial Resistance & Infection Control*, 7(1), 1–7.

Mbuthia, O.W., Mathenge, S.G., Oyaro, M.O. & Ng'ayo, M.O. (2018) Etiology and pathogenicity of bacterial isolates: a cross sectional study among diarrheal children below five years in central regions of Kenya. *Pan African Medical Journal*, 31(1), 88.

Medina, A.M., Rivera, F.P., Pons, M.J., Riveros, M., Gomes, C., Bernal, M. et al. (2015) Comparative analysis of antimicrobial resistance in enterotoxigenic *Escherichia coli* isolates from two paediatric cohort studies in Lima, Peru. *Transactions of the Royal Society of Tropical Medicine and Hygiene*, 109(8), 493–502.

Mentzer, A., Tobias, J., Wiklund, G., Nordqvist, S., Aslett, M., Dougan, G. et al. (2017) Identification and characterization of the novel colonization factor CS30 based on whole genome sequencing in enterotoxigenic *Escherichia coli* (ETEC). *Scientific Reports*, 7(1), 1–11.

Moharana, S.S., Panda, R.K., Dash, M., Chayani, N., Bokade, P., Pati, S. et al. (2019) Etiology of childhood diarrhoea among under five children and molecular analysis of antibiotic resistance in isolated enteric bacterial pathogens from a tertiary care hospital, Eastern Odisha, India. *BMC Infectious Diseases*, 19(1), 1–9.

Nair, G., Ramamurthy, T., Bhattacharya, M., Krishnan, T., Ganguly, S., Saha, D. et al. (2010) Emerging trends in the etiology of enteric pathogens as evidenced from an active surveillance of hospitalized diarrhoeal patients in Kolkata, India. *Gut Pathogens*, 2(1), 4.

Nazarian, S., Gargari, S.L.M., Rasooli, I., Alerasol, M., Bagheri, S. & Alipoor, S.D. (2014) Prevalent phenotypic and genotypic profile of enterotoxigenic *Escherichia coli* among Iranian children. *Japanese Journal of Infectious Diseases*, 67(2), 78–85.

Patel, S.K., Dotson, J., Allen, K.P. & Fleckenstein, J.M. (2004) Identification and molecular characterization of EatA, an autotransporter protein of enterotoxigenic *Escherichia coli*. *Infection and Immunity*, 72(3), 1786–1794.

Puiporm, O., Chantaroj, S., Gangnonngiw, W., Okada, K., Honda, T., Taniguchi, T. et al. (2010) Identification of colonization factors of enterotoxigenic *Escherichia coli* with PCR-based technique. *Epidemiology and Infection*, 138(4), 519–524.

Qadri, F., Svennerholm, A.-M., Faruque, A.S.G. & Sack, R.B. (2005) Enterotoxigenic *Escherichia coli* in developing countries: epidemiology, microbiology, clinical features, treatment, and prevention. *Clinical Microbiology Reviews*, 18(3), 465–483.

Rahman, M.M., Ahmed, P., Kar, A., Sakib, N., Shibly, A.Z., Zohora, F.T. et al. (2020) Prevalence, antimicrobial resistance, and pathogenic potential of enterotoxigenic and enteropathogenic *Escherichia coli* associated with acute diarrheal patients in Tangail, Bangladesh. *Foodborne Pathogens and Disease*, 17(7), 434–439.

Rao, M.C. (1985). Toxins which activate guanylate cyclase: heat-stable enterotoxins. Ciba Foundation Symposium 112 – Microbial Toxins and Diarrhoeal Disease, 112, 74–93.

Rodas, C., Mamani, R., Blanco, J., Blanco, J.E., Wiklund, G., Svennerholm, A.-M. et al. (2011) Enterotoxins, colonization factors, serotypes and antimicrobial resistance of enterotoxigenic *Escherichia coli* (ETEC) strains isolated from hospitalized children with diarrhea in Bolivia. *Brazilian Journal of Infectious Diseases*, 15(2), 132–137.

Roy, K., Hamilton, D., Ostmann, M.M. & Fleckenstein, J.M. (2009) Vaccination with EtpA glycoprotein or flagellin protects against colonization with enterotoxigenic *Escherichia coli* in a murine model. *Vaccine*, 27(34), 4601–4608.

Sabui, S., Debnath, A., Ghosal, A., Wajima, T., Hamabata, T., Ramamurthy, T. et al. (2016) Characterization of oligomeric assembly of colonization factor CS6 from enterotoxigenic *Escherichia coli*. *Microbiology*, 162(1), 72–83.

Shaheen, H.I., Khalil, S.B., Rao, M.R., Abu Elyazeed, R., Wierzba, T.F., Peruski, L.F. et al. (2004) Phenotypic profiles of enterotoxigenic *Escherichia coli* associated with early childhood diarrhea in rural Egypt. *Journal of Clinical Microbiology*, 42(12), 5588–5595.

Simuyandi, M., Chilengi, R., B Connor, S., B. Voeglein, J., M. Laban, N., Mwila-Kazimbaya, K. et al. (2019) Enterotoxigenic *Escherichia Coli* toxins and colonization factors among zambian children presenting with moderate to severe diarrhea to selected health facilities. *Archives of Microbiology & Immunology*, 3(4), 173–184.

Singh, P., Metgud, S.C., Roy, S. & Purwar, S. (2019) Evolution of diarrheagenic *Escherichia coli* pathotypes in India. *Journal of Laboratory Physicians*, 11(4), 346–351.

Subekti, D.S., Lesmana, M., Tjaniadi, P., Machpud, N., Sriwati, Sukarma et al. (2003) Prevalence of enterotoxigenic *Escherichia coli* (ETEC) in hospitalized acute diarrhea patients in Denpasar, Bali, Indonesia. *Diagnostic Microbiology and Infectious Disease*, 47(2), 399–405.

Taha, Z.M. & Yassin, N.A. (2019) Prevalence of diarrheagenic *Escherichia coli* in animal products in Duhok province, Iraq. *Iranian Journal of Veterinary Research*, 20(4), 255–262.

Torres, O.R., González, W., Lemus, O., Pratdesaba, R.A., Matute, J.A., Wiklund, G. et al. (2014) Toxins and virulence factors of enterotoxigenic *Escherichia coli* associated with strains isolated from indigenous children and international visitors to a rural community in Guatemala. *Epidemiology and Infection*, 143(8), 1662–1671.

Vidal, R.M., Muhsen, K., Tennant, S.M., Svennerholm, A.-M., Sow, S.O., Sur, D. et al. (2019) Colonization factors among enterotoxigenic *Escherichia coli* isolates from children with moderate-to-severe diarrhea and from matched controls in the Global Enteric Multicenter Study (GEMS). *PLoS Neglected Tropical Diseases*, 13(1), e0007037.

Vilchez, S., Becker-Dreps, S., Amaya, E., Perez, C., Paniagua, M., Reyes, D. et al. (2014) Characterization of enterotoxigenic *Escherichia coli* strains isolated from Nicaraguan children in hospital, primary care and community settings. *Journal of Medical Microbiology*, 63(Pt_5), 729–734.

Wolf, M.K. (1997) Occurrence, distribution, and associations of O and H serogroups, colonization factor antigens, and toxins of enterotoxigenic *Escherichia coli*. *Clinical Microbiology Reviews*, 10(4), 569–584.

SUPPORTING INFORMATION

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